



Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S) & LIIT

JEE-MAIN

Date: 19-12-2025

Time: 09.00Am to 12.00Pm

GTM-02

Max. Marks: 300

KEY SHEET

MATHEMATICS

1	2	2	4	3	2	4	3	5	1
6	1	7	2	8	2	9	1	10	1
11	1	12	2	13	2	14	1	15	3
16	1	17	2	18	4	19	2	20	4
21	506	22	4	23	2	24	6	25	11

PHYSICS

26	1	27	1	28	1	29	3	30	3
31	1	32	2	33	4	34	2	35	4
36	2	37	3	38	3	39	2	40	4
41	4	42	1	43	1	44	4	45	2
49	1726	47	100	48	24	49	183	50	180

CHEMISTRY

51	4	52	2	53	1	54	2	55	2
56	3	57	4	58	2	59	2	60	1
61	3	62	2	63	4	64	2	65	3
66	2	67	1	68	1	69	3	70	2
71	7	72	50	73	29	74	13	75	17



SOLUTION

MATHEMATICS

$$1. \quad g(x) = \begin{cases} 2+x, & 0 \leq x \leq 1 \\ 2-x, & 1 < x \leq 2 \\ 4-x, & 2 < x \leq 3 \end{cases} \Rightarrow \text{rang} = [0, 3]$$

$$2. \quad f(x) = \sqrt[5]{x^2|x|^3} - \sqrt[3]{x^2|x|} - 1$$

$$\Rightarrow f(x) = \begin{cases} (x^5)^{\frac{1}{5}} - (x^3)^{\frac{1}{3}} - 1, & x \geq 0 \\ (-x^5)^{\frac{1}{5}} - (-x^3)^{\frac{1}{3}} - 1 & x < 0 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} x-x-1, & x \geq 0 \\ -x+x-1 & x < 0 \end{cases} \Rightarrow = -1 \text{ for all } x$$

Thus, $f(x)$ is differentiable every where.

$$3. \quad \bar{a} + \bar{b} + \bar{c} = \bar{0} \Rightarrow \bar{a} + \bar{c} = -\bar{b}$$

$$(\bar{a} + \bar{c}) \times \bar{b} = -\bar{b} \times \bar{b} \Rightarrow \bar{a} \times \bar{b} + \bar{c} \times \bar{b} = \bar{0}$$

$$|(\bar{a} + \bar{b}) + (\bar{c} + \bar{b})| = 0$$

$$\text{From I} \Rightarrow |\bar{c}| = 6(2\sqrt{2} - 1)$$

$$\Rightarrow |\bar{c}| = 6 - 12\sqrt{2} < 0 \text{ (not possible)}$$

$$\Rightarrow \bar{b} \cdot \bar{c} = 12 \Rightarrow \bar{b} \cdot (-\bar{a} - \bar{b}) = 12$$

$$\Rightarrow \bar{b} \cdot (-\bar{a}) - |\bar{b}|^2 = 12 \Rightarrow \bar{b} \cdot (-\bar{a}) = |\bar{b}|^2 + 12 \Rightarrow 24$$

$$\Rightarrow |\bar{b}|^2 - |\bar{a}| \cos \angle ACB = 24 \Rightarrow \cos \angle ACB = \sqrt{\frac{2}{3}}$$

$$4. \quad \frac{P_{25} - P_{23}}{P_{24}} = (-3) \quad \& \quad \frac{Q_{25} + \sqrt{3}O_{24}}{2.Q_{23}} = 8$$

$$5. \quad \sum_{k=1}^{12} a_{2k-1} = -\frac{72}{5}a_1$$

$$a_1 + a_3 + \dots + a_{23} = -\frac{72}{5}a_1 \Rightarrow a + a + 2d + \dots + a + 22d = -\frac{72}{5}a$$

$$\Rightarrow 12a + 2d(1 + 2 + \dots + 11) = -\frac{72}{5}a \Rightarrow 12a + 2d\left(\frac{11 \times 12}{2}\right) = -\frac{72}{5}a$$

$$\Rightarrow 132d = -\frac{132}{5}a \Rightarrow a = -5d \dots \dots \dots (i)$$



$$\text{Also } \sum_{k=1}^n a_k = 0 \Rightarrow S_n = 0 \Rightarrow \frac{n}{2}[2a + (n-1)d] = 0$$

$$\Rightarrow 2a = -(n-1)d \dots \dots (ii)$$

From equation (i) and (ii)

$$(n-1)d = 10d \quad \therefore n = 11$$

6. $C_3 \rightarrow C_3 - C_1$

$$y(x) = \begin{vmatrix} \sin x & \cos x & 1 + \cos x \\ 27 & 28 & 0 \\ 1 & 1 & 0 \end{vmatrix} \Rightarrow y(x) = -(1 + \cos x)$$

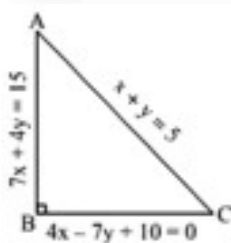
$$\Rightarrow \frac{dy}{dx} = \sin x \Rightarrow \frac{d^2y}{dx^2} = \cos x \Rightarrow \frac{d^2y}{dx^2} + y = -1$$

7. $\Delta = \begin{vmatrix} 3 & 1 & \beta \\ 2 & \alpha & -1 \\ 1 & 2 & 1 \end{vmatrix} = 0 \Rightarrow 3\alpha + 4\beta - \alpha\beta + 3 = 0$

$$\Rightarrow \Delta_3 = \begin{vmatrix} 3 & 1 & \beta \\ 2 & \alpha & -1 \\ 1 & 2 & 1 \end{vmatrix} = 0 \Rightarrow 9\alpha + 19 = 0$$

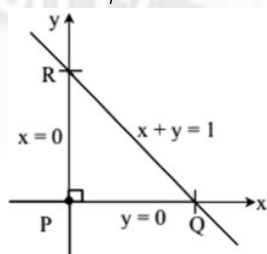
$$\Rightarrow \alpha = \frac{-19}{9}, \quad \beta = \frac{6}{11} \Rightarrow 22\beta - 9\alpha = 31$$

8.



Since triangle is right angle so
Orthocentre will be at right angle vertex

$$\left. \begin{array}{l} 7x + 4y = 15 \\ 7x + 7y + 10 = 0 \end{array} \right\} P.O.I.B = (1, 2)$$



Distance between P and B = $\sqrt{5}$

9. $\operatorname{cosec} \theta = \frac{2(\sqrt{3}-1) \pm \sqrt{4(3+1-2\sqrt{3})+16\sqrt{3}}}{2\sqrt{3}}$

$$= \frac{2(\sqrt{3}-1) \pm \sqrt{16+8\sqrt{3}}}{2\sqrt{3}} \Rightarrow = \frac{2(\sqrt{3}-1) \pm (2+2\sqrt{3})}{2\sqrt{3}}$$



$$\Rightarrow \operatorname{cosec} \theta = 2 \text{ or } \frac{-2}{\sqrt{3}} \Rightarrow \therefore \sin \theta = \frac{1}{2} \text{ or } \frac{-\sqrt{3}}{2}$$

$$\therefore \sin \theta = \frac{1}{2} \text{ has 3 solutions \& also } \sin \theta = \frac{-\sqrt{3}}{2}$$

$$\text{Has 3 solutions in } \left[\frac{-7\pi}{6}, \frac{4\pi}{3} \right]$$

10.

$$\begin{aligned} & \cot^{-1} \left(\frac{|\sec 2| - 1}{\tan 2} \right) - \cot^{-1} \left(\frac{\left| \sec \frac{1}{2} \right| + 1}{\tan \frac{1}{2}} \right) \\ &= \cot^{-1} \left(\frac{-1 - \cos 2}{\sin 2} \right) - \cot^{-1} \left(\frac{1 + \cos \frac{1}{2}}{\sin \frac{1}{2}} \right) = \pi - \cot^{-1}(\cot 1) - \cot^{-1} \left(\cot \frac{1}{4} \right) \\ &\Rightarrow \pi - 1 - \frac{1}{4} = \pi - \frac{5}{4} \end{aligned}$$

11.

$$\beta = \frac{\alpha}{2}, \quad \gamma = \frac{\alpha}{2}$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1 \Rightarrow \cos^2 \alpha + 2 \cos^2 \frac{\alpha}{2} = 1$$

$$\Rightarrow \cos \alpha (\cos \alpha + 1) = 0 \Rightarrow \cos \alpha = 0, -1 \quad \cos \alpha = \frac{\pi}{2}, \pi$$

$$\text{Now } \beta = \frac{\alpha}{2} \Rightarrow \frac{\pi}{4}, \frac{\pi}{2}$$

$$\text{So sum is } \frac{3\pi}{4}$$

12.

$$A(3, \alpha, 3) \quad \& \quad B(-3, -7, \beta)$$

$$\overrightarrow{BA} = 6\hat{i} + (\alpha + 7)\hat{j} + (3 - \beta)\hat{k}$$

$$\vec{p} \times \vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 1 \\ -3 & 2 & 4 \end{vmatrix} \Rightarrow \frac{\left| \overrightarrow{BA} \cdot (\vec{p} \times \vec{q}) \right|}{\left| \vec{p} \times \vec{q} \right|} = 3\sqrt{30}$$

$$\Rightarrow 36 + 15(\alpha + 7) - 3(3 - \beta) = (3\sqrt{30})^2 \Rightarrow 36 + 15\alpha + 105 - 9 + 3\beta = 270$$

$$\Rightarrow 15\alpha + 3\beta = 138 \Rightarrow 5\alpha + \beta = 46$$

13.

$$x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$$

$x = 0$ is not the root of this equation so divide it by x^2

$$x^2 - 3x - 2 + \frac{3}{x} + \frac{1}{x^2} = 0 \Rightarrow x^2 + \frac{1}{x^2} - 2 + 2 - 3\left(x - \frac{1}{x}\right) - 2 = 0$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 - 3\left(x - \frac{1}{x}\right) = 0$$



$$x - \frac{1}{x} = 0$$

$$x^2 - 1 = 0$$

$$x = \pm 1$$

$$\alpha = 1, \beta = -1$$

$$\Rightarrow \alpha^3 + \beta^3 + \gamma^3 + \delta^3$$

$$\Rightarrow = 0 + 3(9 - 3(-1))$$

$$x - \frac{1}{x} = 3$$

$$x^2 - 3x - 1 = 0$$

$$\gamma + \delta = 3$$

$$\gamma\delta = -1$$

$$\Rightarrow 1 - 1 + (\gamma + \delta)((\gamma + \delta)^2 - 3\gamma\delta)$$

$$\Rightarrow = 3(12)$$

$$\Rightarrow = 36$$

$$14. \quad 0 < e^x < 1 \Rightarrow x < 0$$

$$0 < \ln|x| < 1 \Rightarrow 1 < |x| < e \Rightarrow x \in (-e, -1) \cup (1, e)$$

$$\therefore x \in (-e, -1)$$

$$15. \quad 0 \leq \{x\} < 1 \quad (\{x\} \neq 0)$$

$$\sin 0 < \sin \{x\} < \sin 1 \Rightarrow \frac{1}{\sin \{x\}} > 1.18 \Rightarrow \left[\frac{1}{\sin \{x\}} \right] \geq 1$$

$$16. \quad f(x) = 0$$

$$17. \quad \text{area} = ab + \frac{1}{2}(a^2 + b^2)\sin 2\theta$$

$$18. \quad A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$19. \quad n^2 - 25n + 144 = 0 \Rightarrow n = 9 \text{ or } 16$$

Neglecting $n = 16$, since that case largest angle is 195°

20.

$$\vec{r} = k(\vec{b} + \vec{c}) \text{ and } \vec{r} \cdot \vec{a} = 3$$

$$\vec{r} \cdot \vec{a} = k(\vec{b} \cdot \vec{a} + \vec{c} \cdot \vec{a})$$

$$3 = k(\lambda + 6 - 15 + 3 - \lambda + 3\lambda)$$

$$3 = k(3\lambda - 6) \Rightarrow k(\lambda - 2) = 1 \rightarrow 1 \text{ and } |r| = k\sqrt{25 + 4 + (\lambda - 5)^2} = 1$$

$$\Rightarrow \frac{1}{\lambda - 2} \sqrt{\lambda^2 - 10\lambda + 54} = 1 \Rightarrow \lambda^2 - 10\lambda + 54 = \lambda^2 + 4 - 4\lambda$$

$$\Rightarrow 6\lambda = 50$$

$$21. \quad \left| \frac{M(2025) - M(2023)}{M(2021)} \right| = \left| \frac{(\sin^{2025} \theta + \cos^{2025} \theta) - (\sin^{2023} \theta + \cos^{2023} \theta)}{\cos^{2021} \theta + \cos^{2021} \theta} \right|$$

$$= \left| \frac{\sin^{2023} \theta (\sin^2 \theta - 1) + \cos^{2023} \theta (\cos^2 \theta)}{\sin^{2021} \theta + \cos^{2021} \theta} \right| \Rightarrow = \left| \frac{-\sin^{2023} \theta \cos^2 \theta - \cos^{2023} \theta \sin^2 \theta}{\sin^{2021} \theta + \cos^{2021} \theta} \right|$$

$$\left| \frac{-\sin^{2023} \theta \cos^2 \theta (\sin^{2021} \theta + \cos^{2021} \theta)}{\sin^{2021} \theta + \cos^{2021} \theta} \right| = \left| \frac{-\sin^2(2\theta)}{\sin^{2021} \theta + \cos^{2021} \theta} \right| \leq \frac{1}{4}$$



$$\therefore 2024 \left| \frac{M(2025) - M(2023)}{M(2021)} \right| \leq \frac{2024}{4} = 506$$

(or)

$$|M(r) - M(r+2)| = \left| -\frac{1}{4} \sin^2(2\theta) M(r-2) \right|$$

$$\Rightarrow \left| \frac{M(r) - M(r+2)}{M(r-2)} \right| = \left| -\frac{1}{4} \sin^2 2\theta \right| \leq \frac{1}{4} \Rightarrow 2024 \left| \frac{M(2025) - M(2023)}{M(2021)} \right| \leq \frac{2024}{4}$$

22. $S_1 : f(x) = -x^4, \quad S_2 : f(x) = -x^2 + x, \quad S_4 : f(x) = x^3 \quad S_3 : f(x) = (x-1)^4$

23. $|2a-1| = 3[a] + 2\{a\}, \quad \Rightarrow |2a-1| = [a] + 2a$

$$\text{case(i)} : a > \frac{1}{2}$$

$$2a-1 = [a] + 2a$$

$$[a] = -1$$

$$\therefore a \in [-1, 0) \text{ Reject}$$

$$\text{case(i)} : a < \frac{1}{2}$$

$$-2a+1 = [a] + 2a$$

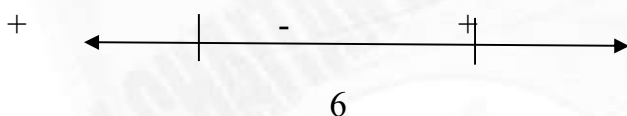
$$a = I + f$$

$$-2(I+f)+1 = I+2I+2f$$

$$I=0, \quad f=\frac{1}{4} \Rightarrow \therefore a=\frac{1}{4}$$

$$\therefore 8 \sum_{a \in S} a = 8 \cdot \frac{1}{4} = 2$$

24. $7x - x^2 - 6 \geq 0, \quad x^2 - 7x + 6 \leq 0, \quad (x-1)(x-6) \leq 0$



$$\therefore x \in [1, 6] \quad \therefore \text{The no. of integer values for which the function is 6}$$

25. Using given conditions;

$$y = (x+1)^2 / 4$$

$$a+1 = \frac{(\alpha+1)^2}{4} \Rightarrow \alpha=3$$

$$m_T = \frac{x+1}{2} = 2 \Rightarrow m_N = -\frac{1}{2} \Rightarrow y-4 = -1/2(x-3)$$



PHYSICS

$$26. \quad KE_R = \frac{1}{2} I \omega^2 = \frac{1}{2} MK^2 \left(\frac{V}{R} \right)^2 = \frac{1}{2} MV^2 \frac{K^2}{R^2}$$

$$KE_R = \frac{50}{100} KE_T = \frac{1}{2} \frac{1}{2} MV^2 \Rightarrow \frac{1}{2} \frac{1}{2} MV^2 = \frac{1}{2} MV^2 \frac{K^2}{R^2} \Rightarrow \frac{K^2}{R^2} = \frac{1}{2}$$

$$\text{In case of solid cylinder } MK^2 = \frac{MR^2}{2}$$

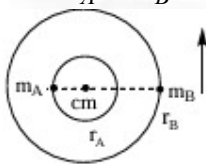
$$27. \quad P = FV$$

$$P = maV = mV \frac{dv}{dt} s$$

$$V dv = \frac{p}{m} dt; \quad \Rightarrow \quad \int_0^v v dv = \frac{p}{m} \int_0^t dt \quad \Rightarrow \quad \frac{V^2}{2} = \frac{p}{m} t$$

$$\Rightarrow v = \sqrt{\frac{2p}{m} t^{\frac{1}{2}}} \Rightarrow \frac{ds}{dt} = \sqrt{\frac{2p}{m} t^{\frac{1}{2}}} \Rightarrow ds = \sqrt{\frac{2p}{m} t^{\frac{1}{2}}} dt \Rightarrow \int_0^s ds = \sqrt{\frac{2p}{m} t^{\frac{1}{2}}} dt$$

$$28. \quad x = \frac{m_B}{m_A + m_B} (r_B + r_A)$$



$$F_G = \frac{Gm_A m_B}{(r_A + r_B)^2} \Rightarrow m_A \times \omega_1^2 = \frac{Gm_A m_B}{(r_A + r_B)^2}$$

$$\frac{m_B}{(m_A + m_B)} (r_A + r_B) \omega_1^2 = \frac{Gm_B}{(r_A + r_B)^2} \Rightarrow \omega_1^2 = \frac{G(m_A + m_B)}{(r_A + r_B)^3}$$

$$\omega_1 = \frac{2\pi}{T_1} \Rightarrow T_1 = \frac{2\pi}{\omega_1} = 2\pi \sqrt{\frac{(r_A + r_B)^3}{G(m_A + m_B)}} \Rightarrow m_B y \omega_2^2 = \frac{Gm_A m_B}{(r_A + r_B)^2}$$

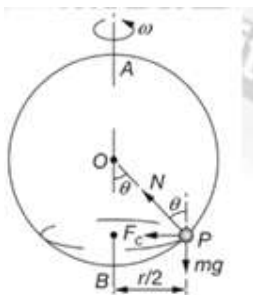
$$\Rightarrow \frac{m_A (r_A + r_B)}{(m_A + m_B)} \omega_2^2 = \frac{Gm_A}{(r_A + r_B)^2} \Rightarrow \omega_2^2 = \frac{G(m_A + m_B)}{(r_A + r_B)^2}$$

$$29. \quad \text{Area under acceleration vs time graph gives velocity}$$

$$v = \int a dt = \int 3 dt = 3t + c$$



30.



Let θ be the angle with the vertical, then $\sin \theta = \frac{r/2}{r} = \frac{1}{2} \Rightarrow \theta = \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$

$$N \cos \theta = mg \Rightarrow N \cos \theta = F_c = \frac{m\omega^2 r}{2}$$

$$\tan \theta = \frac{\omega^2 r}{2g} \Rightarrow \omega^2 = \frac{2g \tan \theta}{r} = \frac{2g}{r\sqrt{3}}$$

31. Since $x = kt^2 = 1.732t^2 \Rightarrow x = \sqrt{3}t^2 \Rightarrow x = 2\sqrt{3}t$

$$\Rightarrow x = a = 2\sqrt{3}ms^{-1} \Rightarrow a = g(\sin 45^\circ - \mu \cos 45^\circ) = 2\sqrt{3}$$

$$\Rightarrow g \frac{1}{\sqrt{2}}(1 - \mu) = 2\sqrt{3} \Rightarrow 1 - \mu = \frac{2\sqrt{6}}{g}$$

$$\Rightarrow \mu = 1 - \frac{2\sqrt{6}}{g} = 1 - \frac{2(2.45)}{9.8} \Rightarrow \mu = 0.5$$

32. Total translational kinetic energy

$$K_r = \frac{3}{2}nRT = \frac{3}{2}pV = 1.5pV$$

Hence, the correct answer is (B)

33. Mass of plate is

$$M = \int_0^R \rho_0 r (2\pi r dr) = \frac{2\pi\rho_0 R^3}{3} \Rightarrow I_C = \int (dm)r^2 = \int_0^R \rho_0 r (2\pi r dr) r^2 = \frac{2\pi\rho_0 R^5}{5}$$

$$\Rightarrow I = I_C + MR^2 = 2\pi\rho_0 R^5 \left(\frac{1}{3} + \frac{1}{5} \right) = \frac{16\pi\rho_0 R^5}{15}$$

$$\Rightarrow I = \frac{8}{5} \left(\frac{2}{3} \pi \rho_0 R^3 \right) R^2 = \frac{8}{5} MR^2$$

34. According to Gauss law for gravitation, we have

$$E(4\pi r^2) = \int \rho_0 4\pi r^2 dr \quad Er^2 = 4\pi G \int_0^r \rho_0 \left(1 - \frac{r^2}{R^2} \right) r^2 dr$$

$$\Rightarrow |E| = E = 4\pi G \rho_0 \left(\frac{r^3}{3} - \frac{r^5}{5R^2} \right)$$

For E to be maximum, $\frac{dE}{dr} = 0 \Rightarrow r = \sqrt{\frac{5}{9}} R$



35. One main scale division, 1 MSD = a cm

36. One Vernier scale division,

$$1 \text{ VSD} = \frac{(n-1) \times (a \text{ cm})}{n}$$

$\therefore$ Least count = 1 MSD – 1 VSD

$$= a - \frac{(n-1)a}{n} = \frac{a(n-n+1)}{n} = \frac{a}{n} \text{ cm} = \frac{10a}{n} \text{ mm} \quad (\because 1 \text{ cm} = 10 \text{ mm})$$

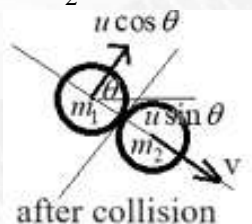
37. The velocity of both bodies m & M are equal. If the block M sticks to the wall, the block m will continue to move which compresses the spring through x. The K.E of the block m will be converted into the potential energy of the spring as it compresses the spring.

Conservation of energy yields

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \Rightarrow x = v\sqrt{\frac{m}{k}}$$

$$\text{Where } \frac{1}{2}(M+m)v^2 = E \Rightarrow v = \sqrt{\frac{2E}{(M+m)}} \quad \therefore x = \sqrt{\frac{2mE}{(M+m)k}}$$

38. By conservation of linear momentum along the line of collision (along the line joining the centers of two spheres), $m_1 u \sin \theta = m_2 v$



Since, Coefficient of restitution for oblique collision, $e = -\frac{v_2 - v_1}{u_2 - u_1}$

$$e = \frac{v}{u \sin \theta} = \frac{m_1}{m_2} = \frac{2}{3}$$

$$39. \quad h = \frac{2T \cos \theta}{r \rho g} \Rightarrow T = \frac{hr \rho g}{2 \cos \theta} \quad \text{or} \quad T \propto \frac{h \rho}{\cos \theta}$$

$$\frac{T_w}{T_{Hg}} = \frac{h_1}{h_2} \times \frac{\cos \theta_2}{\cos \theta_1} \times \frac{\rho_1}{\rho_2} \quad \text{putting the values, we obtain } 1:6.5$$

$$40. \quad \text{Young's modulus } Y = \frac{FL}{Al} = \frac{m4FL}{\pi d^2 l} \Rightarrow = \frac{(4)(1.0 \times 10.0)(2)}{\pi (0.4 \times 10^{-3})^2 (0.8 \times 10^{-3})} \Rightarrow = 2.0 \times 10^{11} \text{ Nm}^{-2}$$

$$\text{Further, } \frac{\Delta Y}{Y} = 2 \left(\frac{\Delta d}{d} \right) + \left(\frac{\Delta l}{l} \right) \quad \therefore \Delta Y = \left\{ 2 \left(\frac{\Delta d}{d} \right) + \left(\frac{\Delta l}{l} \right) \right\} Y \Rightarrow = \left\{ 2 \times \frac{0.01}{0.4} + \frac{0.05}{0.8} \right\} \times 2.0 \times 10^{11}$$

$$= 0.2 \times 10^{11} \text{ Nm}^{-2}$$

$$41. \quad \Delta Q = nC_p \Delta T = 2 \left(\frac{3}{2} R + R \right) \Delta T$$

$$\Rightarrow = 2 \left[\frac{3}{2} R + R \right] \times 5 = 2 \times \frac{5}{2} \times 8.31 \times 5 \Rightarrow = 208 \text{ J}$$

42. A) Dimension of magnetic Induction F = BIL



$$\Rightarrow B = \frac{F}{IL} = \frac{[MLT^{-2}]}{[A][L]} \Rightarrow B = MT^{-2}A^{-1}$$

B) Dimension of magnetic flux = $BA \cos \theta$

$$\Rightarrow B = [MT^{-1}A^{-1}][L^2] = ML^2T^{-2}A^{-1}$$

C) Dimension of magnetic Permeability = $MLT^{-2}A^{-2}$

D) Dimension of magnetization = $M^0L^{-1}A$

43. Spring force is conservative so mechanical energy will be constant. Also average value of kinetic energy & potential energy is constant in a time period.

$$\langle K.E \rangle = \langle P.E \rangle = \frac{1}{2}kA^2$$

44. Given $\Delta Q = 184 \times 10^3 \Rightarrow m = 0.600 \text{ kg at } -12^\circ\text{C} \Rightarrow s = 2222.3 \text{ J/kg}^\circ\text{C}$
And $L = 336 \times 10^3 \text{ J/kg}$

Hence required conversion of ice at -12°C to 0°C is given as

$$Q_1 = 0.600 \times 2222.3 \times 12 = 16000.56 \text{ J}$$

$$\text{Remaining } Q_1 = 184000 - 16000.56 = 167999.44 \text{ J}$$

$$\text{For melting at } 0^\circ\text{C} \Rightarrow Q_2 = 0.600 \times 336000 = 201600 \text{ J needed}$$

$\therefore$ 100% ice is not melted. So final temperature of system is 0°C .

$$\text{Amount of ice melted } 167999.44 = m \times 336000 = 0.4999 \text{ kg}$$

$$\therefore \text{mass of water} = 0.4999 \text{ kg}$$

$$\text{And, mass of ice} = 0.6 - 0.499 = 0.1001 \text{ kg}$$

$$\therefore \text{Ratio} = \frac{0.1001}{0.4999} \approx 1:5$$

45. Mean free path is given by, $\lambda = \frac{RT}{\sqrt{2}\pi d^2 N_A P} \Rightarrow \lambda \propto \frac{1}{d^2}$

$$\text{Kinetic energy is given by, } K = \frac{f}{2} n / RT \Rightarrow K \propto T$$

46. From equation $\vec{v} = \hat{i} + 2\hat{j} \Rightarrow x = t \dots\dots\dots(i)$

$$\Rightarrow y = 2t - \frac{1}{2}(10t^2) \dots\dots\dots(ii)$$

$$\text{From (i) and (ii), } y = 2x - 5x^2$$

47. $W = nRT \ln\left(\frac{V_B}{V_A}\right)$

$$W = 1 \times 8.3 \times 300 \times \ln\left(\frac{4}{2}\right) \Rightarrow = 8.3 \times 300 \times 0.693 = 1725.57$$



$$48. \quad \left| \frac{F}{A} \right| = \eta \frac{dv}{dy} \quad \Rightarrow \quad 10^{-3} = \frac{1}{100} \times \frac{\left(36 \times \frac{5}{18} \right)}{y} y = 100$$

$$49. \quad \text{For a cylinder process, } \Delta U = 0 \quad \Rightarrow \quad Q_{\text{cycle}} = W_{\text{cycle}} \dots\dots\dots(1)$$

$$\text{Where, } W_{\text{cycle}} = W_{A \rightarrow B} + W_{B \rightarrow C} + W_{C \rightarrow A} \dots\dots\dots(2)$$

Process $C \rightarrow A$ is isochoric, hence $W_{C \rightarrow A} = 0$

Process $A \rightarrow B$ isochoric, as its $T - V$ graph is a straight line. Passing through origin. In an isobaric process, work done is $W_{A \rightarrow B} = P(V_B - V_A) = nR(T_B - T_A)$

$$W_{A \rightarrow B} = 1 \times 8.3(400 - 300) = 830 \text{ J} \dots\dots\dots(3)$$

$W_{A \rightarrow B}$ is positive, so expansion of gas takes place. From the equations 1, 2, 3 we get

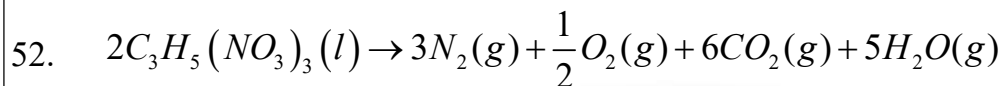
$$-1000 = 830 + W_{B \rightarrow C} + 0 \quad \Rightarrow \quad W_{B \rightarrow C} = -183 \text{ J}$$

$$50. \quad \begin{array}{c} 100^\circ \text{C} \qquad \qquad \qquad 0^\circ \text{C} \\ \hline \Rightarrow \Rightarrow H \end{array} \quad \frac{KA\Delta T}{l} = \frac{mL_{\text{ice}}}{t}$$



CHEMISTRY

51. (a) $N > O > Be > B$ 1st ionisation energy of (N) because of half filled p-orbital.



$$\Delta H_{\text{reaction}}^{\circ} = \sum H_{f(P)}^{\circ} - \sum H_{f(R)}^{\circ}$$

$$\Rightarrow \Delta H_{\text{reaction}}^{\circ} = 3 \times 0 + \frac{1}{2} \times 0 + 6 \times -393.5 + 5 \times -241.8 - (2 \times 364) \Rightarrow = -2361 - 1209 + 728$$

$$\Rightarrow = -2842 \text{ kJ/mol}$$

For 2 moles of nitroglycerine enthalpy -2842 kJ/mol

$\therefore$ For 1 mole of nitroglycerine enthalpy

$$\Rightarrow = \frac{-2842}{2} = -1421 \text{ kJ/mol}$$

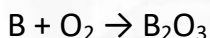
227.1 g of $C_3H_5(NO_3)_3$ has enthalpy -1421 kJ/mol

$$\therefore 10 \text{ g of } C_3H_5(NO_3)_3 \text{ has enthalpy } \Rightarrow = -\frac{1421}{227.1} \times 10 \Rightarrow = 62.5 \text{ kJ/mol}$$

53. More abundant isotope = B^{11}

[Number of neutrons = 6]

$$x = 6$$

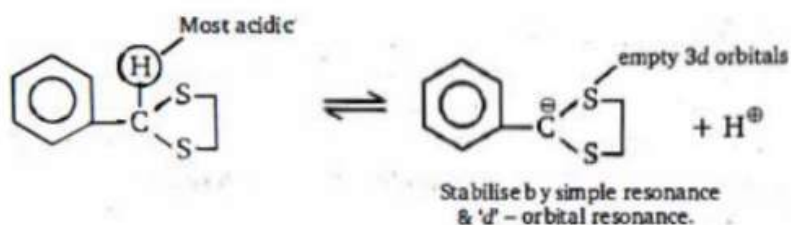


Oxidation state of B in B_2O_3 = +3

So, $y = 3$

Hence $x + y = 9$

54.

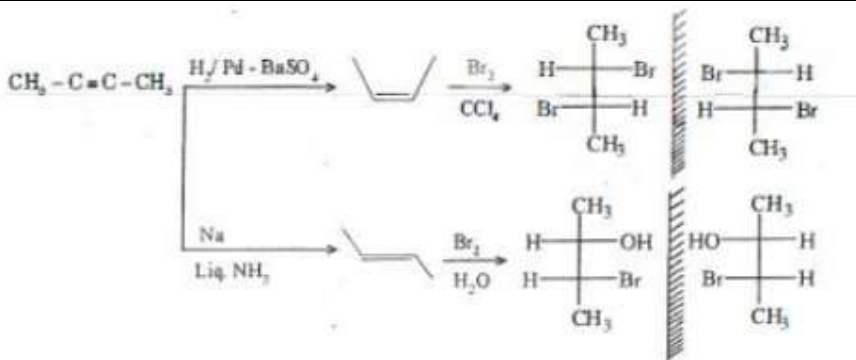


55. $n\lambda = 2\pi r$

$$n\lambda = 2\pi \frac{n^2}{z} a_0 \Rightarrow n\lambda = 2\pi \times \frac{n^2}{Z} \times 52.9 \text{ pm}$$

$$\Rightarrow \lambda = 2\pi \times 52.9 \times 2 \text{ pm} \Rightarrow = 211.6 \text{ pm}$$

56.



57. For one mole of X (7 gm of X), required energy is 720 kJ/mol

So, energy required for 110 mg of

$$X = (720/7) \times 0.110 = 11.3 \text{ kJ}$$

58. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g); \Delta H < 0$

Increasing the temperature lowers equilibrium yield of ammonia. However, at higher temperature the initial rate of forward reaction would be greater than at lower temperature that is why the percentage yield of NH_3 too would be more initially.

59. no. of atoms = Atomicity $\times$ mole $\times N_A$

A) No. of atoms = $6 \times \frac{212}{106} \times N_A = 12N_A$

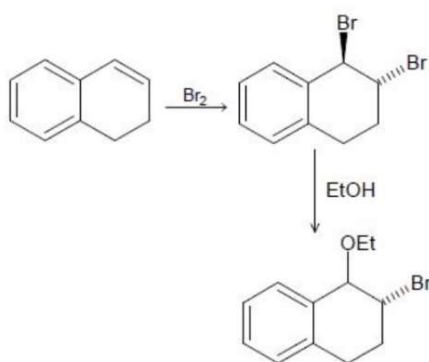
B) No. of atoms = $3 \times \frac{248}{62} \times N_A = 12N_A$

C) No. of atoms = $3 \times \frac{240}{40} \times N_A = 18N_A$

D) No. of atoms = $2 \times \frac{12}{2} \times N_A = 12N_A$

E) No. of atoms = $3 \times \frac{220}{44} \times N_A = 15N_A$

60.



61. In ClF_3 , central atom has 3 bond pairs and 2 lone pair of electrons

62. $H_2O(l) \xrightarrow{1} H_2O(l) \xrightarrow{2} H_2O(s)$
 $(100^\circ C) \quad (0^\circ C) \quad (0^\circ C)$

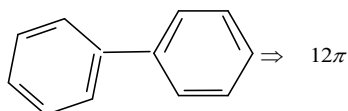
Step-I: $\Delta H_1 = -18 \times 4.2 \times 100 = -7560 \text{ J} = -7.56 \text{ kJ mol}^{-1}$

Step-II: $\Delta H_2 = -6.00 \text{ kJ mol}^{-1}$

$\Delta H = \Delta H_1 + \Delta H_2, \quad \Delta H = \Delta E + \Delta nRT$



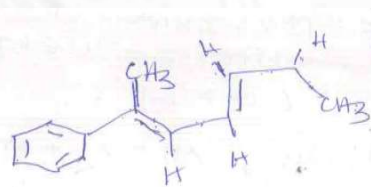
63. Due to inert pair effect; Tl^{+3} and Pb^{-4} can behave as oxidising agents.
64. Higher the number of anti-bonding electrons, lower is the stability
- 65.



Conjugated $\bar{e}$.

66. NCERT statement
67. NCERT statement
68. NCERT statement
69. NCERT statement
70. NCERT statement
71. 1-Hexyne
2-Hexyne
3-Hexyne
3-Methyl 1-pentyne
4-methyl 1-pentyne
4-methyl 2-pentyne
3,3-Dimethyl 1-butyne
72. Order of M.P. of group 14 : $C > Si > Ge > Pb > Sn$
element M.P. ($^{\circ}C$)
 $Z = 6 = C \quad 3730$
 $Z = 14 = Si \quad 1410$
 $Z = 32 = Ge \quad 937$
 $Z = 50 = Sn \quad 232$
 $Z = 82 = Pb \quad 327$

73.



74. $V_B N_B = 10 \times 0.2 \times 2 = 4$
 $V_A N_A = 25 \times 0.1 = 2.5$
 $[OH^-] = \frac{4 - 2.5}{35} = \frac{1.5}{35} = 0.428 = 4.28 \times 10^{-2}$
 $P^{OH} = -\log(4.28 \times 10^{-2}) = 2 - \log 4.28 = 2 - 0.631 = 1.368$
 $P^H = 12.631$
75. $[Ru] = [Kr] 4d^7 5s^1 \quad [Os] = [Xe] 4f^{14} 5d^6 6s^2$